

Retrieval System Lab: RAG Architecture Overview

Retrieval Augmented Generation (RAG) integrates vector search with language models.

Text documents are chunked into smaller passages, typically 300 to 500 characters long.

Each text chunk is converted into a 384-dimensional vector embedding using all-MiniLM-L6-v2.

FAISS Index & Vector Search

FAISS (Facebook AI Similarity Search) is an open-source vector store designed for fast retrieval. Inner Product (IndexFlatIP) combined with L2 normalized embeddings calculates Cosine Similarity. FAISS stores vector indices persistently on disk alongside chunk metadata for instant search.

Android Kotlin & FastAPI Integration

The Android app uses Retrofit 2 with OkHttp to connect to the FastAPI backend at 10.0.2.2:8000.

Android users upload PDF files via multipart/form-data and query the index with top_k = 3.

The user interface follows a clean White and Violet Material 3 design system.